ATHARVA MAHAJAN

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Education

2023 – Present **International Institute of Information Technology, Pune**
Bachelor of Engineering in Computer Engineering (Final Year)
9.1/10

2022 **Sardar G.G. High School, Raver**
12th - HSC Board
87.5%

2020 **Holy Angels' Convent Senior Secondary School, Pune**
10th - SSC Board
93%

Achievements

- **Competitive Programming:** Achieved rating of 1322; highest rank 1918 in Codeforces Round 1032. Id : https://codeforces.com/profile/atharva
- **Intercollegiate Debate Competition:** Participated and qualified for regionals in the competition organised by Times of India and Federal Bank.
- **General Proficiency:** Received the General Proficiency Award for academic excellence for four consecutive years in school academics with merit.

Internship

DSAI Intern — WNS
Worked on the FareAssist – Caching project within the Skense OCR product, developing configurable caching features tailored to client requirements. Contributed to backend development using FastAPI and Python, worked with Docker-based deployments. *Tools:* Python, FastAPI, Docker, Generative AI, Streamlit.

Projects

Redis
Built a lightweight Redis-inspired in-memory key-value server from scratch using native C, with C++ used selectively for implementation utilities. Implemented core operations over RAM-resident data with support for concurrent client requests, key expiration, and low-latency access. Designed the server around Redis-style key-value semantics, focusing on efficient memory management, request processing, and high concurrency. *Tools:* C(Native)

RISC-V Benchmarking Framework
I am currently working on a simulation-based benchmarking framework to evaluate custom RISC-V ISA extensions for accelerating transformer computational kernels on edge devices. The proposed modified RISC-V simulators will compare baseline execution against proposed custom instructions. The goal is to optimise dynamic instruction count and execution cycles. *Tools:* C/C++, Python, Spike/gem5

Mathematical Visualizations using Manim
Created an explainer visualizing Lagrange's Mean Value Theorem (LMVT) and uploaded it to YouTube.
Tools: Python, Manim

Coursework Information

Data Structures and Algorithms — Discrete Mathematics — Computer Graphics — Database Management Systems (DBMS) — Operating Systems — Web Technology — Cloud Computing — Data Science and Big Data Analytics — Combinatorics — Probability — Statistics — Calculus — Artificial Intelligence — Machine Learning

Skills

Programming	C++, JavaScript, ReactJS, Python, Java, MySQL,
Tools	Docker, Git/Github, WordPress, MS Office
Languages	English, Hindi, Marathi, German (B2.2)